

GAJAVADA SANJEEVKUMAR

Ph.D. Scholar, Department of Mechanical Engineering
Indian Institute of Technology (Indian School of Mines), Dhanbad

📍 Plot No. 138, Road No. 5, Sri Ram Nagar Colony, Turkayamjal, Ranga Reddy, Telangana - 501510, India

✉ sanjeevkumargajawada@gmail.com · 📞 (+91) 9030 833 667

🌐 <https://www.linkedin.com/in/skgajawada/> · 🐙 <https://github.com/skgajawada>

🌐 <https://skgajawada.github.io>

Research Objective

To advance mechanical and aerospace engineering through integrated research in structural dynamics, aeroelasticity, vibration control, composite structures, finite element methods (FEM), and computational mechanics. Aiming to bridge physical domain modeling with data-driven techniques—including Machine Learning, Artificial Intelligence, and Federated Learning—to solve complex engineering challenges, develop intelligent mechanical systems, and promote interdisciplinary academic innovation.

Research Interests & Specializations

Core Mechanical & Aerospace Domains

- Aeroelasticity and Flutter Analysis
- Structural Dynamics, Vibrations, and Modal Analysis
- Composite Structures and Advanced Material Modeling
- Finite Element Method (FEM) and Computational Mechanics
- Active Vibration Control & Smart Structures
- Smart Manufacturing and Digital Twins

Data-Driven & Computational Engineering

- Machine Learning & Artificial Intelligence in Mechanical Systems
- Data Science, Engineering Analytics, and Federated Learning for Intelligent Transportation
- Scientific Machine Learning (SciML)
- Structural Optimization

Methodological Focus

- Aeroelastic Analysis of Composite Wings
- Static, Dynamic, and Non-Linear Analysis of Composite Laminates and Plates
- Structural Health Monitoring (SHM)
- Finite Element Formulation & Custom Solver Modeling in MATLAB / ANSYS

Education

Ph.D. in Mechanical Engineering (Design) <i>Indian Institute of Technology (Indian School of Mines), Dhanbad</i>	July 2025 – Present CGPA: 8.20 / 10.0
AICTE QIP Postgraduate Certificate in Data Science <i>Indian Institute of Technology Bhilai, Bhilai</i>	June 2025 – December 2025 CGPA: 8.67 / 10.0
Master of Technology (M.Tech.) in Machine Design <i>GITAM (Deemed to be University), Visakhapatnam</i>	2016 – 2018 CGPA: 8.78 / 10.0
Bachelor of Technology (B.Tech.) in Mechanical Engineering <i>Vivekananda Institute of Technology and Science, Karimnagar</i>	2010 – 2014 Percentage: 71.28%

Intermediate (M.P.C. – Mathematics, Physics, Chemistry) <i>SR Junior College, Karimnagar</i>	2008 – 2010 Percentage: 81.80%
Secondary School Certificate (SSC) <i>Pragathi Vidyalayam, Sircilla</i>	2007 – 2008 Percentage: 85.50%

Professional & Research Experience

Sabudh Foundation (in collaboration with STPI, Mohali) Jan 2026 – June 2026
Online Data Science Intern

- Developed and deployed distributed machine learning models tailored for intelligent transportation systems.
- Implemented privacy-preserving Federated Learning workflows using Python, TensorFlow, Scikit-learn, and Dataiku DSS platforms.

Guru Nanak Institutions Technical Campus, Hyderabad March 2021 – June 2025
Assistant Professor, Department of Mechanical Engineering

- Taught core undergraduate mechanical engineering courses and mentored undergraduate capstone design projects.
- Managed departmental NAAC, NBA, and NIRF accreditation teams and documentation processes.
- Led departmental preparations and compliance tasks during JNTU Hyderabad inspection visits.
- Served as Minor Degree Coordinator, Remedial and Tutorial Classes Coordinator, and Kinematics and Dynamics Laboratory Coordinator.
- Acted as SAE BAJA and Go-Karting Faculty Advisor, implementing Outcome-Based Education (OBE) frameworks.

Sree Chaitanya College of Engineering, Karimnagar Nov 2018 – March 2021
Assistant Professor, Department of Mechanical Engineering

- Instructed undergraduate mechanical engineering subjects and mentored undergraduate research projects.
- Formed and served as Faculty Coordinator for the ISTE Student Chapter.
- Managed operations as Kinematics and Dynamics Laboratory Coordinator and organized state-level technical workshops.

Jyothishmathi Institute of Technology and Science, Karimnagar Dec 2015 – June 2016
Lecturer, Department of Mechanical Engineering

- Conducted lectures and practical laboratory sessions in engineering mechanics, graphics, and workshop practices.
- Served as Academic and Examination Branch Coordinator.

Summary of Teaching Experience

Total Pedagogical Experience: 7+ Years in Undergraduate Engineering Education, Academic Administration, Curriculum Development, Student Mentoring, Laboratory Management, and NBA/NAAC Outcome-Based Education (OBE) Implementation.

Courses Taught / Pedagogical Mastery

- **Computational & Solid Mechanics:** Finite Element Methods (FEM/FEA), Mechanics of Solids, Strength of Materials, Engineering Mechanics.
- **Design & Kinematics:** Design of Machine Members / Elements, Kinematics of Machinery.
- **Thermal & Fluid Sciences:** Thermodynamics, Fluid Mechanics, Heat Transfer.
- **General Mechanical & Industrial Engineering:** Basic Mechanical Engineering, Plant Layout and Material Handling, Operations Research.

- **Graphics & Drawing:** Engineering Graphics, Machine Drawing.

Research & Academic Projects

Data Science Internship Project

Jan 2026 – June 2026

Federated Learning for Intelligent Transportation Systems

- Formulated decentralized, privacy-preserving machine learning frameworks tailored to intelligent transportation applications using Python, TensorFlow, Scikit-learn, Dataiku DSS, and advanced data analytics tools.

IIT Bhilai AICTE QIP PG Project

June 2025 – Dec 2025

AI-Driven Frequency Response Analysis for Transformer Diagnostics via Unified Multi-Format Software

- Developed unified data processing and machine-learning frameworks for automated fault identification in transformer frequency response diagnostics.

Master's Thesis Project (M.Tech.)

2016 – 2018

A Study on the Effect of Elastic-Plastic Material Properties of Metals on Hardness Values Using FEA

- Modeled elastic-plastic behavior of metallic alloys and evaluated theoretical vs. experimental hardness variations.

Bachelor's Thesis Project (B.Tech.)

2013 – 2014

Hydraulic Robotic Arm for Pick and Place Operation

- Designed and fabricated a fluid-driven multi-axis arm for automated material handling operations.

B.Tech Industrial Mini Project: *Study of Internal Combustion Engines & Their Components.*

Technical & Computational Skills

- **Programming Languages:** Python, MATLAB, SQL
- **Engineering Simulation & FEA Tools:** ANSYS Mechanical, ANSYS APDL, MATLAB, Simulink, AutoCAD, SolidWorks
- **Data Science & Machine Learning:** NumPy, Pandas, Scikit-learn, TensorFlow, Keras
- **Data Analytics & Visualization:** Power BI, Tableau, Dataiku DSS, Dataiku Academy Suite, IBM Watson Studio
- **Documentation & Version Control:** LaTeX, Git, GitHub, MS Office Suite (Word, Excel, PowerPoint)

Academic & Administrative Responsibilities

- **Student Coordinator:** D2MMS Summer School, Department of Mechanical Engineering, IIT (ISM) Dhanbad.
- **Departmental Accreditation & Quality Member:** Departmental NAAC, NBA, and NIRF accreditation committee member at GNITC, Hyderabad.
- **JNTUH Inspection Coordinator:** Key departmental member responsible for JNTU Hyderabad inspection documentation and compliance verification at GNITC, Hyderabad.
- **ISTE Student Chapter Founder & Coordinator:** Formed and served as faculty coordinator for the ISTE Student Chapter at Sree Chaitanya College of Engineering, Karimnagar.
- **Academic & Exam Branch Coordinator:** Coordinator for academic scheduling and examination operations at Jyothishmathi Institute of Technology and Science, Karimnagar.
- **Minor Degree Coordinator:** Departmental lead for managing undergraduate Minor Degree curriculums at GNITC, Hyderabad.
- **Tutorial & Remedial Classes Coordinator:** Formulated academic enhancement frameworks and

remedial learning schedules for students.

- **SAE BAJA & Go-Karting Coordinator:** Faculty advisor guiding undergraduate student design, simulation, and racing competition teams at GNITC, Hyderabad.
- **Kinematics & Dynamics Laboratory Coordinator:** In-charge of laboratory setup, equipment maintenance, and experimental modules at GNITC, Hyderabad and Sree Chaitanya College of Engineering, Karimnagar.
- **Undergraduate Student Mentor:** Faculty mentor guiding B.Tech students in academic progression, career development, and capstone projects across multiple institutions.

MOOCs & Professional Certifications

Dataiku Academy

- Certified Generative AI Practitioner · Certified MLOps Practitioner · Certified ML Practitioner
- Certified Developer · Certified Advanced Designer · Certified Core Designer

IBM & Cognitive Class

- Data Literacy Credential · Sensemaking with Data · Exploring Data Credential · Data Fundamentals
- Your Future in Data: Job Landscape · Data Tools & Languages · Clean, Refine & Visualize Data (Watson)
- Data Science in Our World · Introduction to Data Concepts · Python 101 for Data Science

MathWorks

- Control System Design with MATLAB & Simulink · Classical Controller Design & PID Tuning
- Control Design Onramp with Simulink · Simulink Onramp · MATLAB Onramp
- Control System Modeling Essentials · Linearization of Nonlinear Systems

LinkedIn Learning

- Accelerated MATLAB · Python for Data Science and Machine Learning Essential Training Part 1
- Additive Manufacturing: Tips, Tricks, and Techniques · Learning MATLAB
- Learning SQL Programming · MySQL for Non-Programmers · Learning Data Analytics: 1 Foundations
- Six Sigma Foundations · Six Sigma: Green Belt · Operational Excellence Foundations · Statistics Foundations 3

Coursera

- Getting Started with Microsoft Excel
- Business Analytics with Excel: Elementary to Advanced
- From Excel to Power BI

Professional Development

Summer Schools & Advanced Training

- **Data-Driven Modelling for Mechanical Systems (D2MMS)**
Dept. of Mechanical Engineering, IIT (ISM) Dhanbad July 01–05, 2026 (5 Days)

AICTE ATAL Academy Programs

- **Robotics for Industry 4.0: Enabling Smart Manufacturing**
Sri Venkateswara College of Engineering (SVCE), Tirupati Nov 17–22, 2025 (6 Days)
- **Green Technology and Sustainability Engineering**

Faculty Development Programs (FDPs)

- **R&D Excellence through Data Science Case Studies**
Sphoorthy Engineering College, Hyderabad Jun 29–Jul 03, 2026 (5 Days)
- **AI-Driven Business Analytics For Research, Teaching & Industry**
JIET, Jodhpur June 26–30, 2026 (5 Days)
- **AI Powered Teaching Excellence**
Walchand Institute of Technology, Solapur June 22–27, 2026 (6 Days)
- **Next-Generation Mechanical Engineering: Research to Industry**
IEM, Kolkata Jan 16–21, 2026 (6 Days)
- **Recent Advances in Composite Materials**
MVGR College of Engineering, Vizianagaram Nov 17–21, 2025 (5 Days)
- **Two-Week FDP: The Art and Science of Successful Proposal Writing**
CVR College of Engineering, Hyderabad Oct 27–Nov 06, 2025 (11 Days)
- **The Role of Artificial Intelligence in Manufacturing Industries**
MVGR College of Engineering, Vizianagaram Oct 27–31, 2025 (5 Days)
- **Smart Materials and Technologies for Durable Structures (SMTDS'25)**
R.V.R. & J.C. College of Engineering, Guntur Sept 15–19, 2025 (5 Days)
- **Advanced ANSYS Mechanical Software Domain**
Chaitanya Bharathi Institute of Technology (CBIT), Hyderabad Nov 25–29, 2024 (5 Days)
- **Innovations in Robotics and UAVs: Digital Twins & Drone Design**
VNR VJIET, Hyderabad Jul 30–Aug 03, 2024 (5 Days)
- **Two-Week FDP: Transforming Education: Integrating OBE & NEP 2020**
CVR College of Engineering, Hyderabad Jul 22–Aug 02, 2024 (12 Days)
- **Advanced Python Programming**
Guru Nanak Institutions Technical Campus (GNITC), Hyderabad Dec 26, 23–Jan 02, 24 (8 Days)
- **Additive Manufacturing: Present and Future Trends**
VNR VJIET, Hyderabad Nov 01–06, 2021 (6 Days)
- **Additive Manufacturing Processes and Its Applications**
Sri Manakula Vinayagar Engineering College, Puducherry Aug 23–28, 2021 (6 Days)
- **Emerging Trends in Advanced Materials and Manufacturing**
KITS, Warangal July 12–16, 2021 (5 Days)
- **Two-Week FDP: Nano-Composites & Smart Materials in Aerospace**
MLR Institute of Technology, Hyderabad Dec 07–19, 2020 (13 Days)
- **Advanced Welding Technologies**
JAIN (Deemed-to-be University), Bengaluru Aug 04–08, 2020 (5 Days)
- **Solar Energy Recovery and Storage Technologies**
St. Joseph's Engineering College, Mangaluru Aug 03–08, 2020 (6 Days)
- **Engineering Physics and Materials Science**
CBIT, Hyderabad Aug 03–07, 2020 (5 Days)
- **Research Methodologies in Mechanical Engineering**
Narsimha Reddy Engineering College, Hyderabad Jul 30–Aug 04, 2020 (6 Days)
- **Recent Trends in Manufacturing**
Mahatma Gandhi Institute of Technology (MGIT), Hyderabad Jul 28–Aug 01, 2020 (5 Days)
- **Mathematical Modeling & Numerical Techniques 2020**
KITS, Warangal July 27–31, 2020 (5 Days)
- **Nanoscience and Nanotechnology: Current Perspectives**
G. H. Raison College of Engineering, Nagpur Jul 27–Aug 01, 2020 (6 Days)

- **Optimization Techniques for Mechanical Engineers**
Vignan Institute of Technology & Science, Hyderabad Jul 27–Aug 01, 2020 (6 Days)
- **Advanced Composite Materials**
R. R. Institute of Technology, Bengaluru July 23–29, 2020 (7 Days)
- **National Level FDP: Research Opportunities in Advanced Welding**
Hindusthan College of Engineering, Coimbatore July 22–26, 2020 (5 Days)
- **Recent Advances in Renewable Energy & Energy Efficiency**
MGIT, Hyderabad July 21–25, 2020 (5 Days)
- **Lost Foam Castings (LFS-2020)**
Sree Chaitanya College of Engineering, Karimnagar July 20–25, 2020 (6 Days)
- **Digital Image Processing using MATLAB**
Synxa IT Pvt. Ltd., Hyderabad April 24–27, 2020 (4 Days)
- **Frontiers of Research in Mechanical Engineering (FORME 2020)**
SITAM, Vizianagaram 2020 (5 Days)
- **Three-Week Faculty Induction Training Program (PMMMNMST, MHRD)**
Teaching Learning Centre (TLC), IIT Hyderabad May 20–Jun 08, 2019 (20 Days)

Short-Term Courses (STC) & STTPs

- **Enduring Trends in Hydraulic Control Systems: Past, Present & Future**
Indian Institute of Technology (IIT) Madras, Chennai March 22–27, 2021 (6 Days)
- **Design of Experiments in Engineering**
B. V. Raju Institute of Technology (BVRIT), Hyderabad Nov 16–21, 2020 (6 Days)
- **QIP STC: Application of Forecasting Methods in Eng. & Business**
Indian Institute of Technology (IIT) Kharagpur, Kharagpur Nov 12–16, 2020 (5 Days)
- **Recent Advances in MEMS, Mechatronics and Their Applications**
MGIT, Hyderabad Sept 21–26, 2020 (6 Days)
- **Advances in Additive Manufacturing**
YCCE, Nagpur / Siemens CoE, Nagpur Sept 21–26, 2020 (6 Days)
- **Recent Advances in Tribology and Surface Engineering**
Saintgits College of Engineering, Kottayam Sept 14–19, 2020 (6 Days)
- **Advances in Finite Element Method for Industry & Research**
Vasavi College of Engineering, Hyderabad Sept 07–12, 2020 (6 Days)
- **Modeling and Analysis using MATLAB and Python**
VNR VJIET, Hyderabad Aug 03–08, 2020 (6 Days)
- **Teaching Learning Pedagogies**
A. G. Patil Polytechnic Institute, Solapur Aug 03–08, 2020 (6 Days)
- **QIP STC: Theory of Plasticity and Its Applications**
Indian Institute of Technology (IIT) Bombay, Mumbai Nov 25–29, 2019 (5 Days)
- **Green Manufacturing**
NITTTR Chandigarh at SCCE, Karimnagar 5 Days
- **Training of Trainers (TOT) for Teacher Educator**
NITTTR, Bhopal 5 Days
- **Advanced Techniques in Modelling and Analysis**
VNR VJIET, Hyderabad 5 Days

Workshops, Training & Symposia

- **Problem-Driven AI: Real World Applications & Solution Frameworks**
National Institute of Technology (NIT) Rourkela, Rourkela Feb 07–16, 2026 (10 Days)
- **Implementation of OBE and Online Teaching**
NITTTR, Chennai June 21–25, 2021 (5 Days)

- **Emerging Trends in Thermal Engineering**
Sree Chaitanya College of Engineering, Karimnagar May 03–07, 2021 (5 Days)
- **2nd International Conference on “Fluids Under Confinement”**
IIT Kharagpur (SPARC Program), Kharagpur March 14–31, 2021 (18 Days)
- **Metallurgy for Mechanical Engineering**
Sree Chaitanya College of Engineering, Karimnagar Sept 28–Oct 02, 2020 (5 Days)
- **Arduino Workshop (e-Tempesta)**
NSHM Knowledge Campus, Durgapur Jul 27–Aug 02, 2020 (7 Days)
- **National Webinar Series: A Vision Towards Materials & Manufacturing**
KPR Institute of Engineering, Coimbatore July 20–25, 2020 (6 Days)
- **International Webinar: Essential Technologies in Mech. Eng.**
Universal Engineering College, Thrissur 5 Days
- **Autodesk Authorized Academic Partner Event**
GNITC, Hyderabad 5 Days
- **Computational Research Techniques Using MATLAB**
Sree Chaitanya College of Engineering, Karimnagar 5 Days

Honors, Awards & Academic Achievements

National Selections:	Provisionally selected for Ph.D. admission at IIT (ISM) Dhanbad , IIT Mandi , and IIT Dharwad (2025).
Doctoral Fellowship:	Indian Institute of Technology (Indian School of Mines), Dhanbad.
Academic Merit:	Awarded CGPA of 8.67 / 10.0 (IIT Bhilai) and CGPA of 8.78 / 10.0 (GITAM University).
Competitions:	Essay & Elocution Award (Dept. of Youth Services, Karimnagar); Innovation 2K12 Fest Certificate (VITS Karimnagar).

Social Responsibilities & Volunteer Work

- **Blood Donation Volunteer:** Donated blood on 10+ occasions through recognized blood donation drives supporting community healthcare.
- **Pledge Participant:** “Nasha Mukta Bharat, Khushaal Bharat” (Ministry of Social Justice and Empowerment, Govt. of India) – Active advocate for substance abuse awareness and community health.
- **Academic & Student Welfare Volunteer:** Active mentor for student development, social outreach, and welfare activities.

Personal Details

Full Name	: Gajavada Sanjeevkumar
Father's Name	: Rajesham
Date of Birth	: August 01, 1993
Gender	: Male
Nationality	: Indian
Languages Known	: English, Telugu, Hindi

Academic References

Dr. Prashanta Kr. Mahato

Professor

Department of Mechanical Engineering

Indian Institute of Technology (ISM)

Dhanbad, Jharkhand, India

Email: pkmahato@iitism.ac.in

Dr. Rajeswara Reddy Resapu

Professor & Head

Department of Mechanical Engineering

GITAM (Deemed to be University)

Visakhapatnam, India

Email: rresapu@gitam.edu

Dr. Suparno Bhattacharyya

Assistant Professor

Department of Mechanical Engineering

Indian Institute of Technology (ISM)

Dhanbad, Jharkhand, India

Email: suparno@iitism.ac.in

Dr. A. Raj Kumar

Professor & Head

Department of Mechanical Engineering

Guru Nanak Institutions Technical Campus

Hyderabad, India

Email: hodme.gnitc@gniindia.org